

MLDM Assignment 10

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MLDM Assignment 10

This are your test results

Knowledge Questions

☰ Activation function

Status

Seen but not answered

Your score

0 / 10

0%

Response

Choose the correct option for the following statements.

Note :

Correct answer : +2 points

Unanswered : 0 point

Incorrect answer : -1 point

Unanswered	Right	Wrong	
☐	☐	☐	While using the ReLU activation function, all values for $x < 0$ are 0 and they do not respond to variations in input or error as the gradient is 0.
☐	☐	☐	Leaky ReLU does not help in overcoming the dying ReLU problem
☐	☐	☐	The Softmax activation function is usually used in the hidden layer and it is scale-invariant.
☐	☐	☐	The Mixture Density network assumes that any general distribution can be broken into a mixture of normal distributions
☐	☐	☐	Backpropagation cannot be used while a Linear activation function is used on the layers, as the derivative of a linear function will be a constant and hence it is not affected by the input.

☰ Loss function

Status Not started

Your score 0 / 10 0%

Response
Choose the correct option for the following statements.

Note :
Correct answer : +2 points **Unanswered : 0 point**
Incorrect answer : -1 point

Unanswered	Right	Wrong	
☐	☐	☐	A high value for entropy in a probability distribution means that there is high certainty in the distribution.
☐	☐	☐	In Binary Cross-Entropy, the probability that the sample belongs to a negative class is denoted by $(1-p)$, where p is the probability that the sample belongs to the positive class.
☐	☐	☐	Logistic Regression optimizes the binary cross-entropy loss for all the samples on which it is trained.
☐	☐	☐	Cross entropy minimization is an optimization and probability estimation technique, it cannot be used in rare event probability estimation.
☐	☐	☐	Multinomial Distribution models the outcome of an event when performed n times, where the outcome is a discrete value for each trial.

☰ Essay

Status Not started

Your score 0 / 10 0%

Response
Explain in your own words the following activation functions w.r.t Gradient descent (two sentence each).
a. ReLU
b. Softmax
c. Sigmoid

0 word



MLE

Status

Not started

Your score

0 / 10

0%

Response

Probability Mass Function of a discrete random variable X is given below.

X	0	1	2	3
P(X)	$2\theta/3$	$\theta/3$	$2(1 - \theta/3)$	$(1 - \theta/3)$

We took 10 samples from such a distribution, which are independent of each other.

The samples = (0,2,1,2,3,1,3,0,1,2)

Calculate the Maximum Likelihood estimate (MLE) of θ . Hint: Use the Log-likelihood function.

Likelihood of the sample, $L(\theta) = \prod_{i=1}^b P(X_i | \theta)$ is the product of the following terms.

Specify the exponents of each term while calculating using the above formula.

$(2\theta/3)$: has exponent

$(\theta/3)$: has exponent

$(2(1-\theta/3))$: has exponent

$((1-\theta/3))$: has exponent

MLE of θ :

Binary cross entropy

Status

Not started

Your score

0 / 15

0%

Response

Given below is the table containing classification results of 2 classes: apples (1) and peaches(0).

Actual label	Predicted probability	Corrected Probability
0	0.16	
0	0.34	
1	0.41	
0	0.51	
1	0.65	
1	0.74	
0	0.68	
1	0.78	
1	0.84	
1	0.89	

'Actual label' holds the actual class to which the fruit belongs to and 'Predicted probability' holds the probability that the fruit belongs to the apple class (1).

In the given table, fill the values for 'Corrected Probability'

Give your answer with 2 decimal values.

Neural network

Status

Not started

Your score

0 / 15

0%

Response

Consider a neural network with 3-3-1 architecture. 3 input nodes, 3 hidden neurons in 1 layer, and 1 output neuron. This network consists of 2-dimensional input with 1 node for bias term in both layers.

Given ,

The weights from the input layer to the hidden layer, $\alpha = \begin{bmatrix} 1 & 0.4 & 0.3 \\ 0.3 & -1 & 0.5 \end{bmatrix}$

The weights from the hidden layer to the output layer, $\beta = [0.3 \quad 0.2 \quad 0.4]\beta=[0.30.20.4]$

Compute the output of the network for input of (0.7,0.5), using a ReLU activation function at both layers.

Input vector X = [, ,]

$g(h_1) = [$
]

$f(g(h_1)) = [$
]

$g(out) =$

Output of the network : (Give your answer with 3 decimal values.)

Programming

Download Assignment10.ipynb and use it as template for your solution. The template starts with the generation of a list of random numbers between -0.5 and 1.55 as input data. The input data is transformed into two kinds of target data by the application of formula_1 and formula_2. Input and corresponding target data are called dataset, thus we have two datasets. We want to train a neural network once to imitate formula_1 and another time to imitate formula_2. The goal is to observe how the parameters within the neural network are transformed through the training process.

Below the dataset creation in the template, you may find an incomplete class implementation of a minimal neural network, called NeuralNetwork, consisting of 2 layers and with Sigmoid as activation function of the hidden layer, otherwise assume identity as activation function. Below the implementation of the neural network class there is defined a function execute_nn which instantiates the neural network class and shall train it over 100 epochs. Thus, a dataset is used 100 times to train the neural network. After the training, the parameters (weights and bias) of the neural network instance should be printed and the error curve over the epochs should be plotted in logarithmic scale. Furthermore, the target data and the data as outputted by the final neural network is shown in a scatter plot. Since the input data is generated randomly, you may expect different plots for every execution of the notebook. The whole function is called twice, one time for each dataset.

Task instructions are given as comments in the template. Read them carefully and perform the specified tasks.

Upload the .ipynb after completion. Marks will not be given without File upload.

Do not import any new libraries that are not already imported in the template.

 Loss function in neural network

Status

Not started

Your score

0 / 30

0%

Response

Download Assignment10.ipynb and use it as template for your solution. The template starts with the generation of a list of random numbers between -0.5 and 1.55 as input data. The input data is transformed into two kinds of target data by the application of formula_1 and formula_2. Input and corresponding target data are called dataset, thus we have two datasets. We want to train a neural network once to imitate formula_1 and another time to imitate formula_2. The goal is to observe how the parameters within the neural network are transformed through the training process.

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File

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